

Project Proposal Document

Telegram Animation Downloader

(Automation Bridge)

1. Project Team

- Laxman Singh (UID: 25mcc20029)
- Ubaid Khan (UID: 25mcc20014)

2. Objectives of Project

- To bridge the media ecosystem gap between open platforms (Telegram) and closed-ecosystem platforms (WhatsApp) by enabling seamless cross-platform media sharing.
- To eliminate the tedious, manual file-conversion workflow for users migrating custom animated and static sticker packs.
- To engineer a fully automated, asynchronous pipeline that ingests URLs, extracts metadata, and outputs universally compliant media formats (e.g., rigidly scaled 512×512 WebP files).

3. Project Description

This project is a specialized, desktop-based software utility engineered using a robust three-layer architecture designed to handle heavy media processing without locking up the user system:

Presentation Layer

Features a Java Swing graphical user interface that safely isolates all heavy I/O tasks to background worker threads, keeping the application highly responsive and preventing UI freezes.

Network Ingestion Layer

Utilizes Java's asynchronous `HttpClient` and regex-based parsing to communicate securely with the Telegram Bot API and fetch binary `.webp` or `.webm` streams directly to local storage.

Subprocess Orchestration Layer

Programmatically launches an external FFmpeg process to handle complex video

transcoding. It enforces strict resolution scaling and proactive system buffer management (discarding stdout/stderr streams) to prevent system deadlocks during high-load processing.

4. Technologies Used

- **Core Language:** Java (JDK 11+)
- **User Interface:** Java Swing (AWT / JFrame)
- **Networking:** `java.net.http.HttpClient`
- **Data Parsing:** Native Java Regular Expressions (`java.util.regex`)
- **Media Engine:** FFmpeg Command-Line Tool